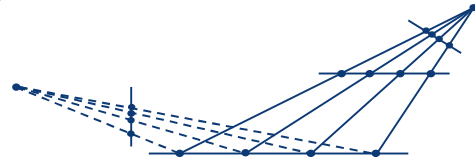


Last time

- Elementary Projective Geometry
 1. Vanishing point
 2. Algebraic manipulation for projective geometry
 3. Projective transformations
 4. Cross Ratio



1

Last time

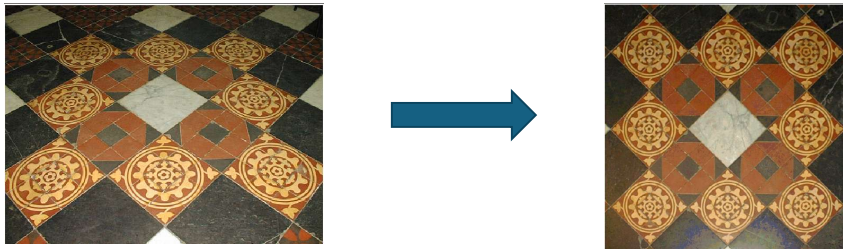
- Elementary Projective Geometry
 1. Vanishing point
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2

Un-distortion (aka Rectification)

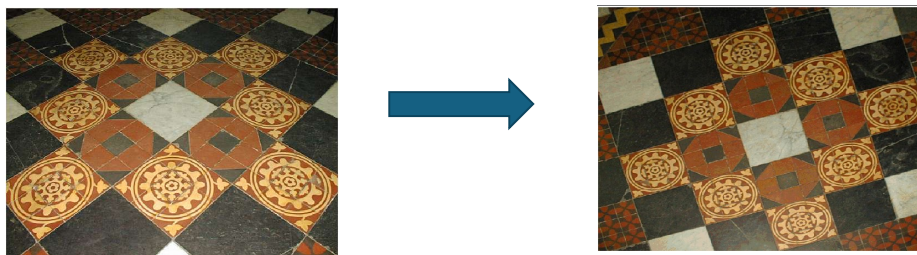
- Un-distortion was achieved by finding the matrix H
 - Un-distortion: going from an unknown projective transform to a similarity transform
- However, this required, potentially, manual correspondence of points



3

Un-distortion (aka Rectification)

- Can we discover a transformation automatically?



4

Composition

- A projective transformation can be decomposed uniquely into a chain of transformations
 - $A = sRK + tv^T$
 - K is upper triangle
 - Unique if s is positive

$$H = \begin{bmatrix} A & \mathbf{t} \\ \mathbf{v}^T & v \end{bmatrix} = H_s H_a H_p = \begin{bmatrix} sR & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} K & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} I & \mathbf{0} \\ \mathbf{v}^T & \alpha \end{bmatrix}$$

5

Agenda

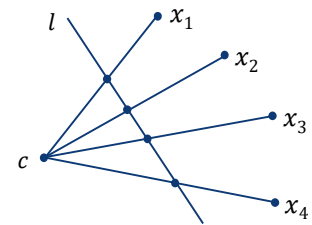
- Four concepts
 - Vanishing point
 - Algebraic manipulation
 - Formulae in projective geometry can sometimes be simpler
 - Projective transformations
 - Cross Ratio

6

Cross Ratio

- Given four points $[x_i, x_j]$ in P^1 the cross ratio is C
 - Definition valid even if one of the points is “ideal”
 - The value does not depend on the “homogeneous constant” (second item in tuple)
- Value invariant to any projective transformation

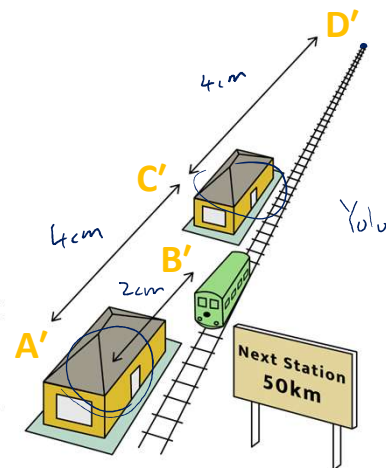
$$C(x_1, x_2, x_3, x_4) \equiv \frac{|x_1 x_2| |x_3 x_4|}{|x_1 x_3| |x_2 x_4|}$$



7

Surveillance camera

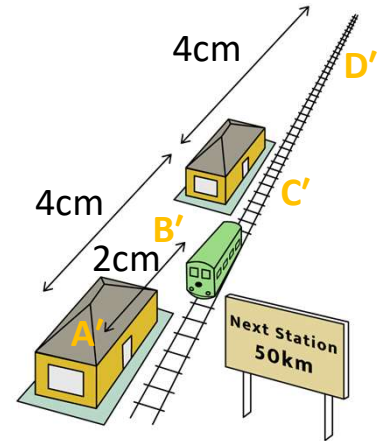
- How far is the train from the next station in the real world?
- We are allowed to do “image processing” and we find pixel distances in the scene
 - The tracks converge in the point D'
 - Distance between the two stations is 4cm
 - Distance between the next station and point D' is 4cm
 - Distance between the train and the previous station is 2cm



9

Surveillance Camera Output

- How far is the train from the next station in the real world?
- We are allowed to do “image processing” and we find that in the picture
 - The tracks converge in the point D’
 - Distance between the two stations is 4cm
 - Distance between the next station and point D’ is 4cm
 - Distance between the train and the previous station is 2cm



10

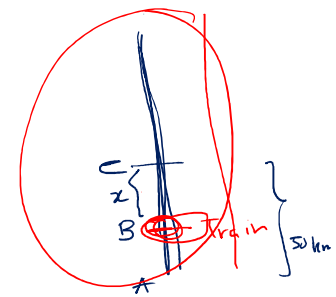
Apply the Cross Ratio Invariant

$$C = \frac{AB \cdot CD}{AC \cdot BD} = \frac{(50-x) \cdot \infty}{50 \cdot \infty}$$

$$C = \frac{(A'B')(C'D')}{(A'C')(B'D')} = \frac{2 \times 4}{4 \times 6} = \frac{1}{3} \stackrel{\Delta}{=} \frac{50-x}{50}$$

Theory

$$\begin{aligned} 50 &= 150 - 3x \\ 3x &= 150 - 50 \\ x &= \frac{100}{3} \approx 33 \text{ km} \end{aligned}$$



11

Using One Known Distance

- How far is the train from the next station?
- Because we work in projective space, the cross ratio is still valid if one of the points is at infinity

- Cross ratio in the real world:

$$\frac{AB * CD}{AC * BD} = \frac{(50 - x) * \infty}{50 * \infty}$$

- Cross ratio in the image is known

$$\frac{A'B' * C'D'}{A'C' * B'D'} = \frac{2 \times 4}{4 \times 6}$$

Ans: 100/3

